

# Python Programming for Beginners

## Day 1: Introduction to Python

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### Learning Objectives

By the end of this lesson, students will be able to:

- Understand what Python is.
  - Explain the uses of Python.
  - Install Python successfully.
  - Write and run their first Python program.
  - Use the `print()` function to display output.
  - Identify common errors in Python programs.
- 

### Why Learn Programming?

Programming is the process of giving instructions to a computer to perform specific tasks. It is an essential skill in today's digital world.

- Benefits of Learning Programming
- Improves logical thinking and problem-solving skills.
- Helps create websites, apps, and software.
- Automates repetitive tasks.
- Used in AI, Machine Learning, Data Science, and Cybersecurity.
- Provides excellent career opportunities in the IT industry.

Key Takeaway :

Learning programming enables you to solve real-world problems, build useful applications, and prepare for a successful career in technology.

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### What is Programming?

Programming is the process of writing a set of instructions that tells a computer how to perform a specific task. These instructions are written in a programming language such as

Python, C, or Java.

Example

A calculator app, a website, or a mobile game works because programmers write code that tells the computer what to do.

Key Points

- Programming gives instructions to a computer.
- It is used to solve problems and automate tasks.
- Programs are written using programming languages.
- It is the foundation of software and application development.

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# History of Python

Python is a high-level, interpreted programming language created by Guido van Rossum. Development of Python began in the late 1980s, and its first official version, Python 0.9.0, was released in 1991.

Guido van Rossum wanted to create a programming language that was simple, easy to read, and powerful. He named it Python after the British comedy show "Monty Python's Flying Circus", not after the snake.

Over the years, Python has become one of the most popular programming languages due to its simple syntax and wide range of applications.

Python Timeline

## Python Timeline

| Year    | Event                                                                                                        |
|---------|--------------------------------------------------------------------------------------------------------------|
| 1989    | Guido van Rossum started developing Python.                                                                  |
| 1991    | Python 0.9.0 was officially released.                                                                        |
| 2000    | Python 2.0 introduced features such as garbage collection.                                                   |
| 2008    | Python 3.0 was released with major improvements.                                                             |
| Present | Python is widely used in AI, Machine Learning, Data Science, Web Development, Cybersecurity, and Automation. |

Key Facts

| Property             | Value                                    |
|----------------------|------------------------------------------|
| Creator              | Guido van Rossum                         |
| First Released       | 1991                                     |
| Latest Major Version | Python 3.x                               |
| Language Type        | High-level, Interpreted, Object-Oriented |
| Inspired By          | ABC Programming Language                 |
| Named After          | Monty Python's Flying Circus             |

### Why Python Became Popular

- Simple and readable syntax
- Easy to learn for beginners
- Free and open source
- Large community support
- Extensive standard library
- Used by companies like Google, Netflix, Instagram, Spotify, NASA, and Microsoft

### Key Takeaway

Python has evolved from a simple programming language into one of the world's most widely used technologies for software development, Artificial Intelligence, Machine Learning, Data Science, Automation, and Cybersecurity.

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## Why Python?

Python is one of the most popular programming languages in the world. It is simple, easy to learn, and powerful. Python is widely used by beginners, students, developers, and large organizations to build websites, automate tasks, analyze data, develop Artificial Intelligence (AI) applications, and much more.

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## Why Choose Python?

### 1. Easy to Learn

Python has a simple and readable syntax, making it one of the best programming languages for beginners.

#### Example:

```
print("Hello, World!")
```

## 2. Easy to Read and Write

Python code is clean and resembles the English language, making it easier to understand and maintain.

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## 3. Open Source

Python is free to download, use, and modify. Anyone can contribute to its development.

Official Website: <https://www.python.org>

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## 4. Cross-Platform

Python programs can run on different operating systems such as:

- Windows
- Linux
- macOS

without changing the code.

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## 5. Large Standard Library

Python provides thousands of built-in modules and libraries that simplify programming.

Examples:

- `math` – Mathematical operations
  - `random` – Generate random numbers
  - `datetime` – Work with dates and time
  - `os` – Interact with the operating system
- 

## 6. Used in Modern Technologies

Python is widely used in many advanced technologies, including:

- Artificial Intelligence (AI)
- Machine Learning (ML)
- Data Science
- Cybersecurity
- Cloud Computing
- Automation

- Internet of Things (IoT)
- 

## 7. Strong Community Support

Millions of developers use Python worldwide. If you face any problem, you can easily find tutorials, documentation, and community support.

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## 8. High Demand in Industry

Python is one of the most in-demand programming languages and is used by companies around the world.

Some companies using Python include:

- Google
  - Netflix
  - Instagram
  - Spotify
  - NASA
  - Microsoft
- 

## Advantages of Python

| Advantage          | Description                        |
|--------------------|------------------------------------|
| Easy to Learn      | Beginner-friendly syntax           |
| Free & Open Source | No license required                |
| Portable           | Runs on multiple operating systems |
| Large Library      | Thousands of built-in modules      |
| Fast Development   | Fewer lines of code                |
| Versatile          | Used in many different fields      |
| Community Support  | Large developer community          |

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## Key Takeaway

Python is a powerful, versatile, and beginner-friendly programming language. Its simple syntax, extensive libraries, and wide range of applications make it an excellent choice for learning programming and building real-world software.

# Applications of Python

Python is a versatile programming language used in various industries due to its simplicity, flexibility, and powerful libraries. It is one of the most widely used programming languages in the world.

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## 1. Web Development

Python is used to develop dynamic and secure websites and web applications.

### Popular Frameworks:

- Django
- Flask
- FastAPI

### Examples:

- Instagram
  - Spotify
  - Pinterest
- 

## 2. Artificial Intelligence (AI)

Python is widely used to build intelligent systems that can think, learn, and make decisions.

### Examples:

- ChatGPT
- Voice Assistants
- Face Recognition
- Recommendation Systems

### Popular Libraries:

- TensorFlow
  - PyTorch
  - Scikit-learn
- 

## 3. Machine Learning (ML)

Machine Learning enables computers to learn from data without being explicitly programmed.

**Applications:**

- Spam Detection
  - Stock Price Prediction
  - Weather Forecasting
  - Fraud Detection
- 

## 4. Data Science

Python is one of the most popular languages for analyzing, processing, and visualizing data.

**Popular Libraries:**

- NumPy
- Pandas
- Matplotlib
- Seaborn

**Applications:**

- Data Analysis
  - Business Reports
  - Data Visualization
- 

## 5. Automation

Python automates repetitive tasks, saving time and reducing human effort.

**Examples:**

- File Management
  - Email Automation
  - Report Generation
  - Web Scraping
- 

## 6. Cybersecurity

Python helps security professionals automate security tasks and analyze vulnerabilities.

**Applications:**

- Network Scanning
  - Password Security Testing
  - Malware Analysis
  - Log Analysis
- 

## 7. Game Development 🎮

Python is used to create 2D games and prototypes.

### **Popular Library:**

- Pygame

### **Examples:**

- Snake Game
  - Tic-Tac-Toe
  - Space Shooter
- 

## 8. Desktop Application Development 💻

Python can be used to build desktop software with graphical user interfaces (GUI).

### **Popular Libraries:**

- Tkinter
- PyQt
- Kivy

### **Examples:**

- Calculator
  - Text Editor
  - Student Management System
- 

## 9. Internet of Things (IoT) 🌐

Python is used to program smart devices and embedded systems.

### **Applications:**

- Smart Home Automation
- Smart Sensors

- Raspberry Pi Projects
- 

## 10. Robotics

Python is widely used in robotics because of its simplicity and extensive libraries.

### Applications:

- Robot Control
  - Path Planning
  - Object Detection
  - Autonomous Robots
- 

## Applications Summary

| Field                   | Examples                           |
|-------------------------|------------------------------------|
| Web Development         | Django, Flask, FastAPI             |
| Artificial Intelligence | ChatGPT, Face Recognition          |
| Machine Learning        | Spam Detection, Prediction Models  |
| Data Science            | Data Analysis, Visualization       |
| Automation              | File & Email Automation            |
| Cybersecurity           | Network Scanning, Malware Analysis |
| Game Development        | Pygame Games                       |
| Desktop Applications    | Calculator, Text Editor            |
| IoT                     | Smart Home, Raspberry Pi           |
| Robotics                | Autonomous Robots                  |

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## Companies Using Python

- Google
- Netflix
- Instagram
- Spotify
- YouTube
- NASA
- Dropbox

- Reddit
  - Microsoft
- 

## Key Takeaway

Python is a powerful and versatile programming language used in web development, Artificial Intelligence, Machine Learning, Data Science, Cybersecurity, Automation, Robotics, IoT, Game Development, and Desktop Application Development. Its simplicity and extensive library support make it one of the best programming languages for beginners and professionals.

## Features of Python

Python is one of the most popular programming languages because of its simplicity, flexibility, and powerful features. It is widely used by beginners and professionals for developing applications in various domains.

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### 1. Simple and Easy to Learn

Python has a simple and readable syntax that is similar to the English language. This makes it easy for beginners to understand and write programs.

**Example:**

```
print("Hello, World!")
```

---

### 2. Interpreted Language

Python is an interpreted language, which means the code is executed line by line without the need for compilation. This makes testing and debugging easier.

**Advantages:**

- Faster program testing
  - Easy debugging
  - No compilation required
- 

### 3. High-Level Language

Python is a high-level programming language that allows programmers to write code without worrying about the hardware or memory management.

**Benefits:**

- Easy to understand
  - Less complex code
  - Faster development
- 

## 4. Object-Oriented Programming (OOP)

Python supports Object-Oriented Programming, allowing developers to organize code using classes and objects.

**Advantages:**

- Code reusability
  - Better organization
  - Easy maintenance
- 

## 5. Cross-Platform (Portable)

Python programs can run on different operating systems without changing the source code.

**Supported Platforms:**

- Windows
  - Linux
  - macOS
- 

## 6. Free and Open Source

Python is free to download and use. Its source code is publicly available, allowing anyone to modify and improve it.

**Official Website:** <https://www.python.org>

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## 7. Large Standard Library

Python comes with a rich collection of built-in libraries and modules that simplify programming.

**Popular Libraries:**

- math
  - random
  - datetime
  - os
  - statistics
- 

## 8. Large Community Support

Python has one of the largest programming communities in the world. Developers can easily find tutorials, documentation, and solutions online.

**Benefits:**

- Free learning resources
  - Community forums
  - Extensive documentation
- 

## 9. Extensible and Embeddable

Python can be integrated with other programming languages such as C and C++ to improve application performance.

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## 10. Dynamically Typed

In Python, you do not need to declare the data type of a variable. Python automatically determines the type during execution.

**Example:**

```
name = "Shivam"  
age = 19  
marks = 92.5
```

---

## 11. Versatile Language

Python is suitable for many types of software development.

**Applications:**

- Web Development

- Artificial Intelligence
  - Machine Learning
  - Data Science
  - Cybersecurity
  - Automation
  - Desktop Applications
  - Game Development
- 

## 12. Fast Development

Python requires fewer lines of code compared to many other programming languages, making software development faster.

### Example:

```
print("Python is Easy")
```

---

## Summary Table

| Feature            | Description                                     |
|--------------------|-------------------------------------------------|
| Simple Syntax      | Easy to read and write                          |
| Interpreted        | Executes code line by line                      |
| High-Level         | Easy for humans to understand                   |
| Object-Oriented    | Supports classes and objects                    |
| Cross-Platform     | Runs on Windows, Linux, and macOS               |
| Free & Open Source | No license required                             |
| Large Library      | Thousands of built-in modules                   |
| Dynamic Typing     | No need to declare variable types               |
| Community Support  | Millions of developers worldwide                |
| Versatile          | Used in AI, Web, Data Science, Automation, etc. |

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## Advantages of Python

- Beginner-friendly
- Easy to learn and use
- Faster software development

- Huge collection of libraries
  - Excellent community support
  - Portable across different operating systems
  - Widely used in industry
- 

## Key Takeaway

Python is a simple, powerful, and versatile programming language. Its easy syntax, large standard library, cross-platform compatibility, and support for modern technologies like Artificial Intelligence, Machine Learning, Data Science, and Cybersecurity make it one of the best programming languages for beginners and professionals.

## Python vs C

Python and C are two popular programming languages, but they differ in syntax, execution, performance, and applications. Python is beginner-friendly and focuses on simplicity, whereas C provides low-level control over the computer's hardware and is known for its speed.

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## Comparison Between Python and C

| Feature           | Python                 | C                        |
|-------------------|------------------------|--------------------------|
| Level             | High-Level Language    | Middle-Level Language    |
| Execution         | Interpreted            | Compiled                 |
| Syntax            | Simple and Easy        | Complex                  |
| Learning          | Beginner Friendly      | Moderate to Difficult    |
| Speed             | Slower                 | Faster                   |
| Memory Management | Automatic              | Manual                   |
| Code Length       | Fewer Lines            | More Lines               |
| Development Time  | Faster                 | Slower                   |
| Error Checking    | Runtime                | Compile Time             |
| Portability       | High                   | High                     |
| Object-Oriented   | Yes                    | No                       |
| Libraries         | Large Standard Library | Limited Standard Library |

| Feature  | Python                                | C                                            |
|----------|---------------------------------------|----------------------------------------------|
| Best For | AI, ML, Web, Data Science, Automation | Operating Systems, Embedded Systems, Drivers |

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## Python Advantages

- Easy to learn and understand.
  - Requires fewer lines of code.
  - Large collection of libraries.
  - Excellent for beginners.
  - Ideal for AI, Machine Learning, Data Science, and Automation.
- 

## C Advantages

- Very fast execution.
  - Efficient memory usage.
  - Suitable for system programming.
  - Provides direct access to hardware.
  - Commonly used in operating systems and embedded systems.
- 

## Example Program

### Python

```
print("Hello, World!")
```

### C

```
#include <stdio.h>

int main()
{
    printf("Hello, World!");
    return 0;
}
```

---

## When to Choose Python?

Choose Python if you want to:

- Learn programming as a beginner.
  - Develop websites.
  - Build AI and Machine Learning projects.
  - Perform Data Analysis.
  - Automate repetitive tasks.
- 

## When to Choose C?

Choose C if you want to:

- Develop operating systems.
  - Build embedded systems.
  - Create device drivers.
  - Learn computer memory and pointers.
  - Write high-performance applications.
- 

## Key Differences

- Python is easier to learn than C.
  - C executes faster than Python.
  - Python automatically manages memory, while C requires manual memory management.
  - Python is mainly used for software development, AI, and Data Science.
  - C is mainly used for system programming and embedded applications.
- 

## Conclusion

Python and C are both powerful programming languages. Python is the best choice for beginners because of its simple syntax and wide range of applications. C is preferred when high performance, memory control, and system-level programming are required. Learning both languages provides a strong foundation for software development and computer science.

## Python vs Java

Python and Java are two of the most popular programming languages in the world. Both are powerful and widely used, but they differ in syntax, execution, performance, and applications.

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# Comparison Between Python and Java

| Feature           | Python                                            | Java                                                   |
|-------------------|---------------------------------------------------|--------------------------------------------------------|
| Level             | High-Level Language                               | High-Level Language                                    |
| Execution         | Interpreted                                       | Compiled to Bytecode and runs on JVM                   |
| Syntax            | Simple and Easy                                   | More Complex                                           |
| Learning          | Beginner Friendly                                 | Moderate                                               |
| Speed             | Slower                                            | Faster                                                 |
| Memory Management | Automatic                                         | Automatic (Garbage Collection)                         |
| Code Length       | Fewer Lines                                       | More Lines                                             |
| Development Time  | Faster                                            | Slower                                                 |
| Object-Oriented   | Supports OOP and Procedural Programming           | Pure Object-Oriented (except primitive types)          |
| Platform          | Cross-Platform                                    | Cross-Platform (JVM)                                   |
| Best For          | AI, ML, Data Science, Automation, Web Development | Enterprise Applications, Android Apps, Banking Systems |

## Python Advantages

- Easy to learn and understand.
- Simple and readable syntax.
- Requires fewer lines of code.
- Large collection of libraries.
- Excellent for AI, Machine Learning, Data Science, and Automation.

## Java Advantages

- High performance.
- Platform independent using the Java Virtual Machine (JVM).
- Strong security features.
- Excellent for large-scale enterprise applications.
- Commonly used for Android application development.

## Example Program

## Python

```
print("Hello, World!")
```

## Java

```
public class Main {  
    public static void main(String[] args) {  
        System.out.println("Hello, World!");  
    }  
}
```

---

## When to Choose Python?

Choose Python if you want to:

- Learn programming as a beginner.
  - Build AI and Machine Learning projects.
  - Perform Data Analysis.
  - Automate repetitive tasks.
  - Develop web applications quickly.
- 

## When to Choose Java?

Choose Java if you want to:

- Develop Android applications.
  - Build enterprise software.
  - Create banking and financial systems.
  - Develop large-scale business applications.
- 

## Key Differences

- Python has a simpler syntax than Java.
  - Python code is shorter and easier to write.
  - Java programs generally execute faster.
  - Python is widely used in AI, Machine Learning, and Data Science.
  - Java is widely used in enterprise software and Android development.
- 

## Conclusion

Python and Java are both powerful programming languages. Python is the best choice for beginners because of its simple syntax and versatility. Java is preferred for enterprise applications, Android development, and projects that require high performance and strong security. Learning both languages provides a solid foundation for software development.

## Installing Python

Before writing Python programs, you need to install Python on your computer. Python is free, open source, and available for Windows, Linux, and macOS.

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### System Requirements

- Windows 10 or Windows 11
  - Minimum 4 GB RAM
  - Internet connection
  - Administrator permission (recommended)
- 

### Step 1: Download Python

1. Open your web browser.
2. Visit the official Python website:

**\*\***<https://www.python.org>**\*\***

3. Click on the **Downloads** menu.
  4. Download the latest stable version of Python for your operating system.
- 

### Step 2: Run the Installer

1. Double-click the downloaded Python installer.
2. The Python Setup window will appear.

### Important

✓ Check the option:

#### Add Python to PATH

This allows you to run Python from the Command Prompt.

Then click:

## Install Now

---

### Step 3: Wait for Installation

The installer will automatically install Python.

After the installation is complete, you will see the message:

**Setup was successful**

Click **Close**.

---

### Step 4: Verify the Installation

Open **Command Prompt**.

Type the following command:

```
cmd
python --version
```

Example Output:

```
Python 3.13.2
```

You can also check using:

```
cmd
python
```

If Python is installed correctly, you will see:

```
Python 3.13.2
```

```
>>>
```

To exit Python, type:

```
exit()
```

or press:

```
Ctrl + Z
```

```
Enter
```

---

### Step 5: Verify pip Installation

`pip` is Python's package manager. It is used to install additional Python libraries.

Type:

```
cmd  
pip --version
```

Example Output:

```
pip 25.x.x from ...
```

---

## Step 6: Install Visual Studio Code (Optional)

1. Download VS Code from:

<https://code.visualstudio.com>

2. Install VS Code.
3. Open VS Code.
4. Install the following extensions:

- Python (Microsoft)
  - Jupyter (Microsoft)
- 

## Common Installation Errors

| Problem                  | Solution                                              |
|--------------------------|-------------------------------------------------------|
| python is not recognized | Reinstall Python and enable <b>Add Python to PATH</b> |
| pip is not recognized    | Reinstall Python or add Python Scripts folder to PATH |
| Installation failed      | Run the installer as Administrator                    |

---

## Installation Checklist

- Downloaded Python
  - Installed Python
  - Enabled **Add Python to PATH**
  - Verified Python version
  - Verified `pip`
  - Installed VS Code (optional)
-

## Key Takeaway

Python installation is simple and takes only a few minutes. After installation, verify Python and `pip` using the Command Prompt. You are then ready to start writing and running Python programs using IDLE, VS Code, or Jupyter Notebook.

## First Python Program

```
In [9]: print('Hello, World!')
```

Hello, World!

## Explanation

- `print()` is a built-in Python function used to display output.
  - `"Hello, World!"` is a **string** enclosed in quotation marks.
  - When the program runs, Python prints the message on the screen.
- 

## Syntax

```
print("message")
```

## More Examples

```
In [ ]: print("My name is Shivam.")  
        print("I am a software developer.")
```

My name is Shivam.

I am a software developer.

## Comments in Python

Comments are notes added to a Python program to explain the code. They are ignored by the Python interpreter and are not executed.

---

## Types of Comments

### 1. Single-Line Comment

A single-line comment starts with the `#` symbol.

**Example:**

```
# This is a single-line comment  
print("Hello, World!")
```

---

## 2. Multi-Line Comment

Python does not have a separate syntax for multi-line comments. Triple quotes ( `'''` or `"""` ) are commonly used to write multi-line descriptions.

**Example:**

```
"""  
This is a  
multi-line comment.  
"""  
  
print("Python")
```

---

You can paste this directly into a Markdown cell.

## 3. Inline Comment

An **inline comment** is written on the same line as a Python statement. It is used to explain a specific part of the code.

### Syntax

```
statement  # Comment
```

### Example

```
print("Hello, World!")  # Prints a greeting message
```

---

## Why Use Comments?

- Explain the purpose of the code.
  - Make the program easier to understand.
  - Help during debugging and maintenance.
  - Improve code readability.
- 

## Key Points

- Single-line comments begin with `#`.
- Multi-line comments can be written using triple quotes (`'''` or `"""`).
- Comments are ignored during program execution.

## Example of Comment :

Single line Comment

```
In [5]: # This is a single-line comment  
print("Hello, World!")
```

Hello, World!

Multi-Line Comment

```
In [4]: """  
This is a  
multi-line comment.  
"""  
print("Python")
```

Python

Inline Comment

```
In [ ]: print("Hello, World!") # Prints a greeting message
```

## Common Errors in Python

Beginners often make mistakes while writing Python programs. These mistakes are called **errors**. Understanding common errors helps in debugging programs.

---

### 1. Syntax Error

A **Syntax Error** occurs when Python code does not follow the correct syntax.

**Example:**

```
print("Hello"
```

**Error:**

SyntaxError: '(' was never closed

---

### 2. Name Error

A **Name Error** occurs when a variable or function is used before it is defined.

**Example:**

```
print(message)
```

**Error:**

NameError: name 'message' is not defined

---

## 3. Indentation Error

An **Indentation Error** occurs when the required indentation is missing or incorrect.

**Example:**

```
if True:  
print("Hello")
```

**Error:**

IndentationError: expected an indented block

---

## 4. Type Error

A **Type Error** occurs when incompatible data types are used together.

**Example:**

```
print("Age: " + 20)
```

**Error:**

TypeError

---

## 5. Zero Division Error

A **ZeroDivisionError** occurs when a number is divided by zero.

**Example:**

```
print(10 / 0)
```

**Error:**

ZeroDivisionError: division by zero

---

## Summary Table

| Error             | Cause                          |
|-------------------|--------------------------------|
| SyntaxError       | Incorrect Python syntax        |
| NameError         | Undefined variable or function |
| IndentationError  | Incorrect indentation          |
| TypeError         | Incompatible data types        |
| ZeroDivisionError | Division by zero               |

---

## Key Points

- Errors are common while learning Python.
- Read the error message carefully.
- Correct the mistake and run the program again.

## Best Practices in Python

Following good programming practices helps you write clean, readable, and error-free code.

---

### 1. Use Meaningful Variable Names

Choose variable names that describe their purpose.

#### Good Example:

```
student_name = "Shivam"  
age = 19
```

---

### 2. Follow Proper Indentation

Python uses indentation to define code blocks. Always use **4 spaces** for indentation.

#### Example:

```
if True:  
    print("Hello")
```

---

### 3. Write Comments

Use comments to explain your code.

**Example:**

```
# Print a welcome message  
print("Welcome")
```

---

## 4. Keep Code Simple

Write simple and easy-to-understand code.

---

## 5. Use Meaningful Output

Display clear and understandable messages.

**Example:**

```
print("Student Record Saved Successfully")
```

---

## Summary

| Practice                  | Benefit                  |
|---------------------------|--------------------------|
| Meaningful Variable Names | Improves readability     |
| Proper Indentation        | Prevents errors          |
| Comments                  | Explains the code        |
| Simple Code               | Easy to understand       |
| Meaningful Output         | Improves user experience |

---

## Key Points

- Write clean and readable code.
- Use proper indentation.
- Add comments where necessary.
- Choose meaningful variable names.
- Keep programs simple and organized.

# Exercises

Write five `print()` statements.

## Summary

In this lesson, we learned the fundamentals of Python programming. We understood what programming is and why Python is a popular programming language. We explored the history, features, and applications of Python, and compared it with C and Java.

We also learned how to install Python, run Python programs, write our first program using the `print()` function, use strings and comments, identify common errors, and follow basic coding best practices.

## Key Takeaways

- Python is a simple, high-level, and beginner-friendly programming language.
- `print()` is used to display output.
- Strings are enclosed in single ( `' '` ) or double ( `" "` ) quotation marks.
- Comments improve code readability and are ignored during execution.
- Python programs can be run using the Python Interpreter, IDLE, VS Code, or Jupyter Notebook.
- Reading error messages helps in identifying and fixing mistakes.
- Writing clean and well-formatted code is a good programming practice.

After completing Day 1, you should be able to install Python, run simple programs, use the `print()` function, and understand the basic structure of a Python program.

## Homework

Install Python and write 10 `print()` programs.